DIVYANSH (DS) SOOD

Software Engineer · Production Systems · ML Infrastructure

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Work Authorization: F-1 STEM OPT, 36 months of U.S. work authorization (E-Verify eligible, no H-1B required until 2029)

EDUCATION

Carnegie Mellon University (CMU), Pittsburgh, PA — Dec 2026
M.S. Software Engineering, Scalable Systems (QPA: **4.13/4.33**)
Relevant Coursework: Distributed Systems, Machine Learning, LLM Systems, Cloud Computing

Shri Govindram Seksaria Institute of Technology and Science (SGSITS), Indore, India — May 2023
B.Tech. Computer Science and Engineering

Algorithms & Competitive Programming: LeetCode Knight **Top 4.5%** (Rating **1894+**, **800+**); Rank **308** / **25,000+** BiWeekly 96; Rank **502** ACM ICPC.

TECHNICAL SKILLS

GPU & ML Systems: PyTorch, CUDA, Huggingface, DeepSpeed ZeRO-3, data-parallel & distributed training, model evaluation, performance profiling & benchmarking

AI Coding Tools: Claude Code, Cursor, GitHub Copilot, Windsurf, used daily to explore code and build fast while validating and refining AI-generated output

Languages: Python, Java 21 (Project Loom), C/C++ (CUDA), JavaScript/TypeScript, SQL, Bash

Backend & Cloud: Docker, Kubernetes (concepts), AWS (EC2, SQS, SNS, Lambda, S3, CloudWatch), PostgreSQL, Redis, Kafka, gRPC, REST, microservices, CI/CD

Practices: monitoring & observability (Prometheus/Grafana), JMeter load profiling & reliability testing, failure recovery, performance optimization, debugging, Git, Agile/Scrum, Linux, Slurm

EXPERIENCE

ZIGNEX (INDIA) PVT. LTD. — **Bangalore, India**
Associate Software Engineer, Backend & Data Infrastructure — *Jul 2023 – Jul 2025*

- Owned backend data infrastructure in production across **1,000+** daily batch jobs; refactored the O(n^4) asset-deduplication algorithm to O(n^2) with composite-key hash-map data structures, eliminating the CPU bottleneck.
- Upgraded Java 11 to Java 21 with Project Loom virtual threads, boosting API throughput **10x** and cutting I/O latency **∼70%**, JMeter-validated under production load.
- Launched the Spring Boot data-ingestion microservice and centralized gateways, standardizing **100,000+** unstructured enterprise assets into data pipelines for downstream AI models, adopted by all clients.

GOSENTRIX, INC. — **Sunnyvale, CA**
Applied AI Engineer (Infra & Runtime Focus) — *May 2026 – Aug 2026*

- Engineered an autonomous scanner-orchestration runtime with **95% precision** / **99% recall** on repository discovery, orchestrating agents via A2A-protocol message passing.
- Integrated a runtime Zero-Trust layer that validates every AI-generated output before downstream consumption, establishing trust in production AI workloads.

IRALOGIX SPONSORED PROJECT (CMU MSE PRACTICUM) — **Pittsburgh, PA**
Forward Deployed Engineer, LLM Evals (MSE Capstone) — *Aug 2026 – Present*

- Designed a centralized MCP gateway connecting external AI agents to core IRA APIs with strict schema enforcement, FINRA Rule 4511 audit-compliant retention, and Auth0/OIDC entitlements.

RESEARCH

Carnegie Mellon University, Language Technologies Institute (LTI) — **Pittsburgh, PA**
Graduate Research Assistant, Prof. Lei Li Lab — *Aug 2026 – Present*

- Developed agent-evaluation harnesses (mini-swe-agent, OpenHands) running GLM models inside per-task Docker sandboxes across **186 real CVE-fix tasks**, with end-to-end cost and trajectory instrumentation.

PROJECTS

From-Scratch LLM Training Stack *CUDA, Python, Slurm; CMU 11-868 · github.com/divyanshsood00/flashattention_on_gpt2*
*Built an LLM systems stack from scratch: rank-general CUDA kernels including fused softmax/layernorm (**12.1x** geomean speedup); a reverse-mode autodiff engine reaching **77%** best-val SST-2; an 8.3M-parameter GPT-2-style decoder at **15.1 BLEU** on IWSLT14 (Slurm H100); hand-rolled data-parallel training in PyTorch scaling **1.956x** on two L40S GPUs with bitwise-verified gradient sync; survived Slurm job deaths losslessly through per-epoch evaluation artifacts, keeping long runs recoverable.*

Scotty: LLM Tool-Calling Agent *TypeScript; CMU 17-643 course team project*
*Extended a course-team ticketing platform with LLM tool calling and function execution: eight schema-validated tools in a bounded agentic loop; the purchase path fenced by JWT identity binding with a pessimistic-write inventory lock; **83** vitest cases gating CI.*